# Mohammad Uzair Khan

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## Education

**BITS Pilani, Dubai Campus** — Dubai, UAE
*B.E. Computer Science* — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

**Bishop Conrad Senior Secondary School** | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

## Profile

Computer Science undergraduate in **AI solutions development**: **AI agents** with typed tool calling, retrieval and natural language inference, that turn pharmaceutical data into **actionable insights** for commercial teams. Self-driven and structured: every system is measured on held-out data against a baseline, with clear reporting of what worked and what did not.

## AI Solutions & Data Science Projects

**Rx Market Pulse: AI analyst agent and forecasting** | *Python, SQL, TypeScript, LLM tool calling* — Live | Code
- Built an **AI agent** answering commercial questions over **36.1M** Medicaid rows through **five typed tools** (spend, growth drivers, territories, forecasts, trials), with validated inputs, provenance and refusals; passed 20/20 curated real-model checks
- Backtested quarterly spend forecasts over **28 quarters** (chosen on 2022–23, tested on 2024–25); reported that the chosen method's 9.01% WAPE lost to the seasonal baseline's **5.54%**

**SignalDesk AI: oncology competitive-intelligence agent** | *Python, SQL, Gemini, TypeScript* — Live | Code
- Bounded **Gemini agent** with five typed evidence tools over **29** FDA announcements, 8 formulations and **30** trial records; citation validation, Redis-backed quotas and exportable briefs
- Paired evaluation against a same-model BM25 baseline on **40 held-out** tasks: agent 17/40 vs baseline 19/40, short of the 34/40 target; published case-level results and failure analysis

**PharmaEvidence: AI claim review for commercial content** | *Python, BM25, DeBERTa NLI, ONNX* — Live | Code
- Checks draft claims against **2,309** passages from versioned KEYTRUDA and WINREVAIR labels with BM25 retrieval, numeric and qualifier rules and a pretrained **INT8 DeBERTa NLI** model running on-device
- Froze **120** authored claims (72 development / 48 held out): NLI **47/48**, rules + NLI **48/48**; the rules caught a tenfold dose error the model accepted

**Agent Desk: two AI agents for GitHub and LinkedIn** | *Python, FastAPI, Claude, Next.js* — Live | Code
- Patch audits repositories and drafts fixes, Pitch drafts posts; rules before the model (12 repositories audited with 30 requests, **0 model calls**), an allow-list of writes and dry-run by default; **469** tests

**Footnote: question answering over company filings** | *Python, BM25, embeddings, LLM agents* — Live | Code
- Measured retrieval on all **150** FinanceBench questions (360 filings, 53,399 pages): hit@10 **44%** with the right filing, 31% with a company/year filter; gold-text search hits **97%**, so the gap is vocabulary

**Also built:** Field Force Planner *(territory optimisation)* · Drug Price Watch · Payer Access · CampaignOps — GitHub

## Experience

**Backend Developer Intern, Amaani (Travel-Tech Startup)** — Dubai, UAE
*BITS Practice School (PS-1)* — *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

## Research & Publications

**Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices** — 2025 – 2026
*First Author; submitted to IEEE MSN 2026 (under review)*
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

## Technical Skills

**AI Solutions & Agents**: AI agents, tool calling, RAG, retrieval (BM25, embeddings), natural language inference, structured output, Claude and Gemini APIs
**Machine Learning**: scikit-learn, TensorFlow/Keras, TensorFlow Lite, INT8 quantization, Transformers.js, ONNX Runtime, held-out evaluation, bootstrap intervals, backtesting
**Programming**: Python (Pandas, NumPy), SQL (DuckDB, MySQL), TypeScript, Java, C, FastAPI, Flask, REST APIs, Next.js/React
**Data & Reporting**: data harmonization, Excel, Power BI (DAX), automated reporting, Git, GitHub Actions, pytest, Playwright

## Certifications

The Ultimate Job Ready Data Science Course – CodeWithHarry (2026)